



# IndianOil

## JOURNEY RISK MANAGEMENT (JRM) STUDY

### Salem Terminal to T.K.S. FUELS STATION

#### Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

#### Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

#### Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

#### Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

## Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

## Environmental Considerations

The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

## Journey Risk Mitigation

The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

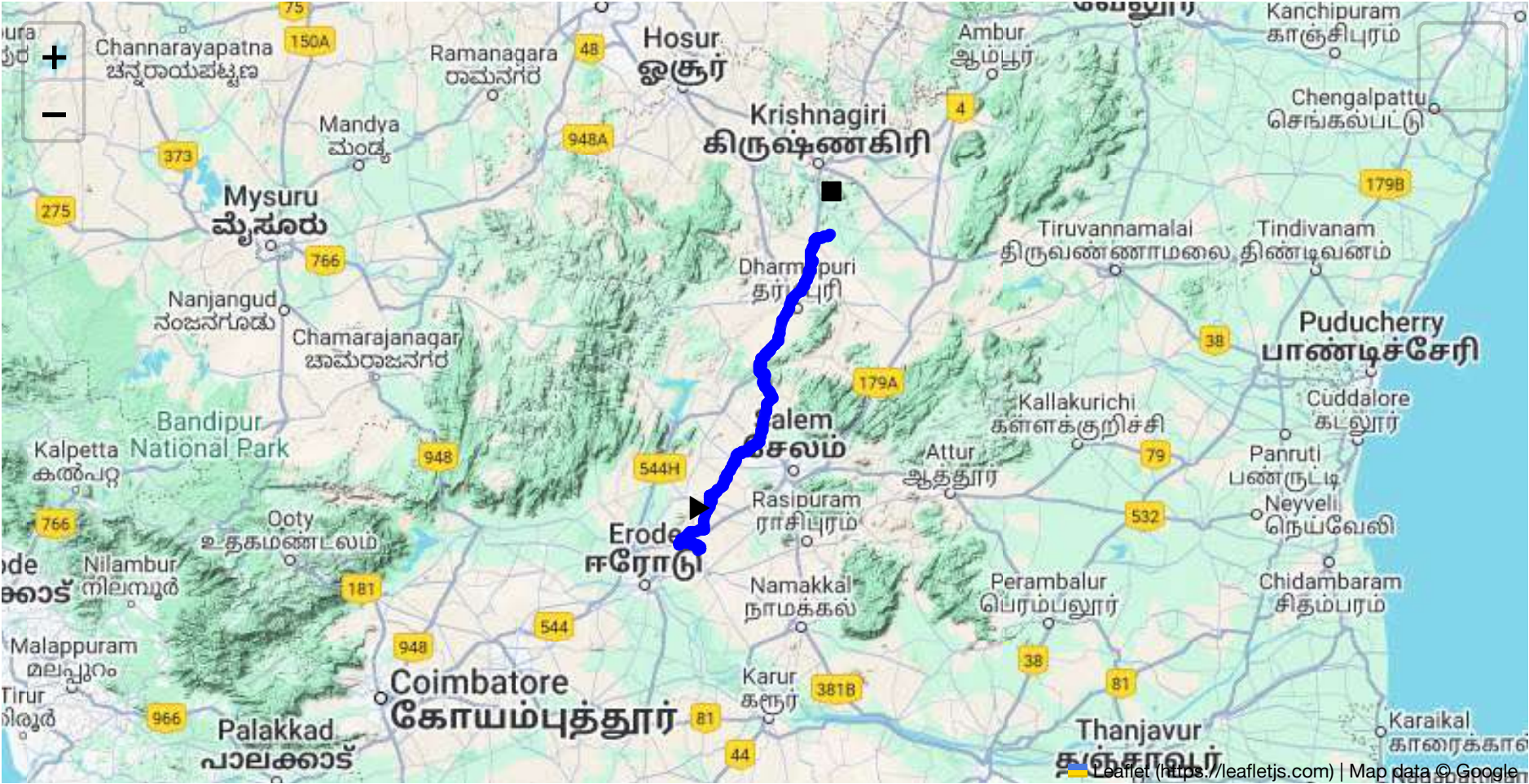
## Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



**Route Summary:**  
**Total Distance: 132.52 km**  
**Estimated Duration: 2.6 hours**  
**Adjusted Duration (Heavy Vehicle): 3.3 hours**

Start: (11.4381, 77.8734)  
End: (12.328031, 78.250788)



## Welcome to the Journey Risk Management Study

### 1. Overview of the Route Map

The route from CVQF+23W, Sangagiri to S No 356/1C3B, Pannandur Rd traverses through varying terrains and several localities in Tamil Nadu. Starting from Sangagiri, the route leads north towards Bhavani Main Road, subsequently connecting through Ayveli in SH 86 and passing through Salem. It further stretches towards Omalur before finally reaching the destination at Thatrahalli. This route primarily covers state highways and involves navigating urban, semi-urban, and rural settings.

### 2. Typical Weather Conditions and Potential Weather-Related Hazards

Tamil Nadu generally experiences a tropical climate with monsoons influencing weather patterns. Between October to December, the region often experiences heavy rainfall due to the northeast monsoon. In such conditions, flooding and reduced visibility might pose significant driving hazards. Summers can be intensely hot, with temperatures rising above 40°C, potentially leading to road surface damage and overheated engines.

### 3. Analysis of Traffic Patterns

Salem, being a major city en route, experiences significant traffic congestion, particularly during peak hours (8-10 AM and 5-8 PM). The roads around Omalur and other small towns may also face congestion during market hours or local events. Rural sections between these urban areas usually have lesser traffic but occasional slow-moving vehicles like tractors or animal-driven carts can cause delays.

### 4. Assessment of Road Quality and Infrastructure

The state highways (SH 86) are generally in good condition but may have sudden patches of deterioration due to seasonal wear. Urban roads in Salem are typically well-maintained, but construction or repair work

might occasionally obstruct traffic. In rural areas, be aware of poorly-lit sections, and be cautious of stray livestock.

## 5. Suggestions for Alternative Routes for Emergencies

In the event of a major blockade or emergency, a viable alternative would be to use the NH44, which runs parallel to some sections of the intended route and offers larger lanes suitable for heavy vehicles. Diversions onto NH544 can also be considered if there are disruptions on the main route near Salem.

## 6. Summary of Local Regulations Affecting Hazardous Material Transport

Tamil Nadu enforces strict regulations on the transportation of hazardous materials, requiring permits and adherence to safety protocols such as prominently displaying necessary signage. Transportation during night hours might be restricted in populated areas to minimize risks.

## 7. Overview of Historical Incidents

There have been few notable incidents involving heavy vehicles on Tamil Nadu's highways, predominantly related to driver error or mechanical failure. No significant accidents involving hazardous materials have been recently reported, but caution is advised in high-traffic zones like Salem.

## 8. Environmental Considerations and Sensitive Areas

The route passes through areas with agricultural activities; thus, accidents involving hazardous materials could adversely affect local farmland. Areas near Salem and some rural sections might also be proximate to environmentally sensitive zones, necessitating cautious driving to avoid ecological disturbance.

## 9. Analysis of Communication Coverage

Cellular coverage is generally reliable along the entire route, although rural stretches might experience brief connectivity issues. It is advisable to carry alternate communication means, especially during emergencies.

## 10. Estimated Emergency Response Times

Response times can vary; in urban parts such as Salem, emergency services can be expected within 15-30 minutes. In more rural areas, it may extend to 45 minutes or more due to less accessible roads.

## 12. Overall Summary of Risk Assessment

In summary, this route presents moderate risks mainly associated with traffic congestion, weather conditions, and rural road quality. Truck drivers should be prepared for potential delays and ensure their vehicles are in top condition to handle varied terrains and climates. Adherence to regulations and clear communication with local authorities is essential when transporting hazardous materials. Environmental sensitivity and community impact should be considered as part of the risk management strategy. Regular updates on road and weather conditions will further mitigate unforeseen risks.

Risk Assessment - Turns

|    | Risk Type  | Risk Level | Coordinates        | Speed Limit |
|----|------------|------------|--------------------|-------------|
| 0  | Turn       | High       | 11.43968, 77.87345 | 15 KM/Hr    |
| 1  | Turn       | High       | 11.44029, 77.87544 | 15 KM/Hr    |
| 2  | Turn       | High       | 11.44900, 77.87406 | 15 KM/Hr    |
| 3  | Turn       | High       | 11.45353, 77.85697 | 15 KM/Hr    |
| 4  | Turn       | Medium     | 11.45781, 77.85203 | 30 KM/Hr    |
| 5  | Turn       | High       | 11.46307, 77.84941 | 15 KM/Hr    |
| 6  | Turn       | High       | 11.45542, 77.81725 | 15 KM/Hr    |
| 7  | Turn       | High       | 11.45605, 77.81683 | 15 KM/Hr    |
| 8  | Turn       | Medium     | 11.49383, 77.88501 | 30 KM/Hr    |
| 9  | Blind Spot | Blind Spot | 11.49400, 77.88538 | 10 KM/Hr    |
| 10 | Turn       | High       | 11.49469, 77.88488 | 15 KM/Hr    |
| 11 | Turn       | High       | 11.55618, 77.89789 | 15 KM/Hr    |
| 12 | Turn       | Medium     | 11.55629, 77.89888 | 30 KM/Hr    |
| 13 | Turn       | High       | 11.58228, 77.90841 | 15 KM/Hr    |
| 14 | Turn       | High       | 11.58340, 77.90637 | 15 KM/Hr    |
| 15 | Turn       | Medium     | 11.59201, 77.90649 | 30 KM/Hr    |
| 16 | Turn       | Medium     | 11.63882, 77.94859 | 30 KM/Hr    |
| 17 | Turn       | Medium     | 11.64161, 77.95177 | 30 KM/Hr    |
| 18 | Turn       | Medium     | 11.64226, 77.95200 | 30 KM/Hr    |
| 19 | Turn       | Medium     | 11.64716, 77.95118 | 30 KM/Hr    |
| 20 | Turn       | Medium     | 11.67868, 77.96836 | 30 KM/Hr    |
| 21 | Turn       | Medium     | 11.70077, 77.97957 | 30 KM/Hr    |
| 22 | Turn       | High       | 11.70223, 77.97860 | 15 KM/Hr    |
| 23 | Turn       | Medium     | 11.71402, 77.99586 | 30 KM/Hr    |
| 24 | Turn       | Medium     | 11.71704, 77.99948 | 30 KM/Hr    |
| 25 | Turn       | Medium     | 11.71918, 78.01127 | 30 KM/Hr    |
| 26 | Turn       | Medium     | 11.73769, 78.03451 | 30 KM/Hr    |
| 27 | Turn       | Medium     | 11.73785, 78.03551 | 30 KM/Hr    |
| 28 | Turn       | High       | 11.73651, 78.03871 | 15 KM/Hr    |
| 29 | Turn       | Medium     | 11.74271, 78.04717 | 30 KM/Hr    |
| 30 | Turn       | Medium     | 11.86763, 78.08336 | 30 KM/Hr    |
| 31 | Turn       | Medium     | 11.87000, 78.08385 | 30 KM/Hr    |
| 32 | Turn       | Medium     | 11.88060, 78.08128 | 30 KM/Hr    |
| 33 | Turn       | Medium     | 11.91329, 78.05861 | 30 KM/Hr    |
| 34 | Turn       | Medium     | 11.92540, 78.06494 | 30 KM/Hr    |
| 35 | Turn       | Medium     | 11.95482, 78.04687 | 30 KM/Hr    |

|    | Risk Type | Risk Level | Coordinates        | Speed Limit |
|----|-----------|------------|--------------------|-------------|
| 36 | Turn      | Medium     | 11.95633, 78.04755 | 30 KM/Hr    |
| 37 | Turn      | High       | 11.95837, 78.04986 | 15 KM/Hr    |
| 38 | Turn      | Medium     | 11.96054, 78.04953 | 30 KM/Hr    |
| 39 | Turn      | High       | 11.96347, 78.05142 | 15 KM/Hr    |
| 40 | Turn      | Medium     | 11.96415, 78.05126 | 30 KM/Hr    |
| 41 | Turn      | Medium     | 11.96519, 78.04958 | 30 KM/Hr    |
| 42 | Turn      | High       | 11.96620, 78.04920 | 15 KM/Hr    |
| 43 | Turn      | Medium     | 11.96842, 78.05097 | 30 KM/Hr    |
| 44 | Turn      | Medium     | 12.24648, 78.19537 | 30 KM/Hr    |
| 45 | Turn      | Medium     | 12.25238, 78.20039 | 30 KM/Hr    |
| 46 | Turn      | High       | 12.31467, 78.20768 | 15 KM/Hr    |
| 47 | Turn      | Medium     | 12.31791, 78.21461 | 30 KM/Hr    |
| 48 | Turn      | High       | 12.31908, 78.21586 | 15 KM/Hr    |
| 49 | Turn      | High       | 12.32377, 78.24215 | 15 KM/Hr    |
| 50 | Turn      | Medium     | 12.32526, 78.24360 | 30 KM/Hr    |

### Emergency Locations

|    | type     | name                              | coordinates            | speed_limit | risk_level |
|----|----------|-----------------------------------|------------------------|-------------|------------|
| 1  | hospital | Sugumar Hospital                  | 11.7396031, 78.0426955 | 30 km/h     | Medium     |
| 2  | hospital | Omalur Government Hospital        | 11.74161, 78.044296    | 30 km/h     | Medium     |
| 3  | clinic   | R.R. Nursing Home                 | 11.7394129, 78.0439861 | 30 km/h     | Medium     |
| 4  | hospital | Sri Hari Hospital                 | 11.7396726, 78.0440701 | 30 km/h     | Medium     |
| 5  | hospital | Sri Sugam Hospital                | 11.7429821, 78.0461039 | 30 km/h     | Medium     |
| 6  | hospital | Saravana Hospital                 | 11.7409303, 78.0458308 | 30 km/h     | Medium     |
| 7  | hospital | Arun Hospital                     | 11.7429892, 78.0466881 | 30 km/h     | Medium     |
| 8  | clinic   | Saratha Clinic                    | 11.7422818, 78.0458175 | 30 km/h     | Medium     |
| 10 | clinic   | Sri Meenakshi Nursing Home, Salem | 11.7449162, 78.0474637 | 30 km/h     | Medium     |
| 11 | clinic   | Divyam Child Care Clinic          | 11.7447135, 78.0473858 | 30 km/h     | Medium     |
| 12 | clinic   | Momota Clinic                     | 11.744982, 78.047397   | 30 km/h     | Medium     |
| 13 | clinic   | Gomathi Clinic                    | 11.7445293, 78.0470401 | 30 km/h     | Medium     |
| 14 | hospital | Vinayaka Child Hospital           | 11.745812, 78.0480225  | 30 km/h     | Medium     |
| 15 | hospital | Dr. G.R.Thilakam Hospital         | 11.746097, 78.04769    | 30 km/h     | Medium     |
| 16 | hospital | Thulasi Hospital                  | 11.7461516, 78.0469984 | 30 km/h     | Medium     |
| 17 | hospital | Amirthalingam Hospital            | 11.7456708, 78.0470548 | 30 km/h     | Medium     |

|    | type     | name                               | coordinates            | speed_limit | risk_level |
|----|----------|------------------------------------|------------------------|-------------|------------|
| 18 | hospital | Children Hospital                  | 11.7458377, 78.0480205 | 30 km/h     | Medium     |
| 19 | clinic   | Shri Ramana Poly Clinic            | 11.7477122, 78.0479293 | 30 km/h     | Medium     |
| 20 | hospital | Dr. Nagaraj Hospital               | 11.7464035, 78.0472401 | 30 km/h     | Medium     |
| 21 | clinic   | Manimuthu Skin Clinic              | 11.7521189, 78.0484058 | 30 km/h     | Medium     |
| 23 | hospital | Anugraha Multi Speciality Hospital | 11.773549, 78.0546408  | 30 km/h     | Medium     |
| 25 | clinic   | JSA Clinic                         | 11.8700619, 78.0844982 | 30 km/h     | Medium     |
| 26 | hospital | L.R. Hospital                      | 11.8703354, 78.0844554 | 30 km/h     | Medium     |
| 27 | hospital | Thoppur,Government Hospital        | 11.9398824, 78.0551658 | 30 km/h     | Medium     |
| 29 | hospital | Government Hospital                | 12.3048002, 78.2060881 | 30 km/h     | Medium     |

Crowded Spots

|    | type    | name                                              | coordinates            | speed_limit | risk_level |
|----|---------|---------------------------------------------------|------------------------|-------------|------------|
| 0  | school  | KRP Matric. Hr. Sec School                        | 11.4546193, 77.8142445 | 30 km/h     | Medium     |
| 9  | school  | Fathima Hr Sec School                             | 11.7410508, 78.0479272 | 30 km/h     | Medium     |
| 22 | school  | John Britto Matriculation Higher Secondary School | 11.7676907, 78.0513092 | 30 km/h     | Medium     |
| 24 | college | Dhirajlal Gandhi College of Technology            | 11.7968014, 78.0562095 | 30 km/h     | Medium     |
| 28 | college | Laxminarayana Polytechnic College                 | 12.0421277, 78.103203  | 30 km/h     | Medium     |
| 30 | school  | Government Boys Higher secondary school           | 12.3047059, 78.2065495 | 30 km/h     | Medium     |

Route Photos of Risky Spots



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.44029, 77.87544



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.44900, 77.87406



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.45353, 77.85697



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.45781, 77.85203



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.46307, 77.84941



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.45542, 77.81725



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.45605, 77.81683



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.49383, 77.88501



**Risk Type:** Blind Spot  
**Risk Level:** Blind Spot  
**Speed Limit:** 10 KM/Hr  
**Coordinates:** 11.49400, 77.88538



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.49469, 77.88488



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.55618, 77.89789



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.58340, 77.90637



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.59201, 77.90649



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.63882, 77.94859



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.64161, 77.95177



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.64226, 77.95200



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.64716, 77.95118



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.67868, 77.96836



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.70077, 77.97957



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.70223, 77.97860



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.71402, 77.99586



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.71704, 77.99948



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.71918, 78.01127



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.73769, 78.03451



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.73785, 78.03551



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.73651, 78.03871



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.74271, 78.04717



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.86763, 78.08336



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.87000, 78.08385



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.88060, 78.08128



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.91329, 78.05861



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.92540, 78.06494



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.95482, 78.04687



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.95633, 78.04755



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.95837, 78.04986



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.96054, 78.04953



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.96347, 78.05142



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.96415, 78.05126



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.96519, 78.04958



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.96620, 78.04920



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.96842, 78.05097



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 12.24648, 78.19537



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 12.25238, 78.20039



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 12.31467, 78.20768



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 12.31791, 78.21461



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 12.31908, 78.21586



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 12.32377, 78.24215

